Supplementary Reading Material: On Dual Spaces

- 阅读提示: 无限维线性空间的对偶空间是一个含糊的概念. 通常的教材上有如下三类处理方式:
  1. (右倾) 仅考虑可构造的对偶空间, 例如 $(\mathbb{R}[x])^* \simeq \mathbb{R}[\![x]\!]$, 但对一般情形避而不谈;
  2. (折中) 明面上不使用甚至不承认选择公理, 但在背地里默认了一些反直觉的公理: 例如默认非零空间存在非零的对偶空间, 但从不给出证明 (补充: 这是似乎课堂采取的方式, 但请务必验证课堂证明之正确性, 以及是否使用了选择公理 (选择公理的等价形式: 任何子空间存在补空间));
  3. (左倾) 直接使用选择公理.

  有探索欲的读者不会接受右倾的处理方式, 而选择公理不在教学范围内. 本节习题将折中派的做法进行公理化, 用一条表述简单的额外公理为教材打上恰当的补丁. 我们最终承认通常公理与额外的 HB 公理.

  - (通常公理). ZF 与 DC, 也就是矩阵论和数学分析承认的东西. 特别地, DC 用于证明可数个可数集的并仍可数, 以及无限维线性空间存在无穷大的线性无关组, 其严格独立于 ZF.
  - (HB 公理). 对任意的子空间 $U \hookrightarrow V$, 映射 $f : U \to W$ 的定义域总能被扩大至 V. 换言之, 总能找到 $\widetilde{f} : V \to W$ 使得 $\widetilde{f}|_U = f$.

  可以发现, 张贤科等教材的公理体系严格等价于通常公理 + HB 公理.

  若不承认 HB, 我们无法排除 Exotic Examples (见习题) 的存在性. 可使用这些例子自查是否伪证了某些命题, 类似地, 有限域的重要作用也是自查伪证与否.

  习题目录: 有限维对偶空间一瞥, 不承认 HB 的后果, 承认 HB 的好处, 对偶空间的 Galois 对应, 对偶空间的所有基本结论.

Finite-Dimensional Vector Spaces and Dual Spaces

Definition. Let V be a vector space over a field $\mathbb{F}$. The dual space is defined as

$$V^* := \mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F}).$$

Example. For $\dim V = n < \infty$, with basis $u_{i1 \le i \le n}$, the dual space V^* is also n-dimensional with basis (dual basis)

$$\{f_i\}_{1 \le i \le n}, \quad f_i : V \to \mathbb{F}, \quad u_j \mapsto \delta_{i,j}.$$

From the perspective of bilinear forms, the pairing

$$V \times V^* \to \mathbb{F}, \quad (u, f) \mapsto f(u)$$

has matrix form $I_{n \times n}$ under the bases $\{u_i\}_{1 \le i \le n}$ and $\{f_i\}_{1 \le i \le n}$.

Note. The isomorphism $V \simeq V^*$ is artificial. An informal but illustrative analogy: assume V has unit kilometre (km), then V^* has unit km^{-1}. The linear isomorphism $V \simeq V^*$ is never natural.

Example. The canonical linear map $\eta_V : V \to V^{**}$ is a natural isomorphism.

- The term "natural" signifies that for any linear map $f : U \to V$, the compositions $U \xrightarrow{f} V \xrightarrow{\eta_V} V^{**}$ and $U \xrightarrow{\eta_U} U^{**} \xrightarrow{f^{**}} V^{**}$ coincide.

Exotic Examples in Linear Algebra Without the Axiom of Choice

Example. ($\dim V = \infty$, yet $V^* = 0$). *If the axiom of choice is not assumed*, then there do exist infinite-dimensional vector spaces V with dual space $V^* = 0$. The example by H. Läuchli presents an infinite-dimensional vector space V whose proper subspaces are all finite-dimensional.

Example. (When V^* is strictly larger than V, yet $V \simeq V^{**}$). *In ZF set theory with dependent choice*, it takes effort to verify $\mathbb{R}[[x]]^* \simeq \mathbb{R}[x]$. However, $\mathbb{R}[[x]]$ does not possess a countable basis.

Proposition. The canonical linear map $V \to V^{**}$ is injective if and only if V is a subspace of some product space $\prod_X \mathbb{F}$. For instance, $\mathbb{F}[x]$ and $\mathbb{F}[[x]]$ both embed into their double dual spaces.

Proof. See this article.

Axiom of Extension (HB)

Axiom. (Axiom of Extension, also known as the Hahn-Banach Axiom (HB)).

- For any linear subspace $U \subseteq V$, every linear map $f : U \to W$ extends to a map $\widetilde{f} : V \to W$ such that $\widetilde{f}|_U = f$.

Remark. The Hahn-Banach Axiom is independent of the usual axioms (ZF + DC), which are assumed in the study of tensor products of infinite-dimensional vector spaces.

Exercise. The Hahn-Banach Axiom is equivalent to the assertion that $\mathrm{Hom}_{\mathbb{F}}(-, W)$ always maps injections to surjections.

Exercise. By HB, for any $v_1, v_2 \in V$ with $v_1 \neq v_2$, there exists $f \in V^*$ such that $f(v_1) \neq f(v_2)$. Hence, the canonical linear map $V \to V^{**}$ is always injective.

Exercise. By HB, show that for $U \hookrightarrow V$, the following map is an isomorphism

$$\frac{V^{**}}{U^{**}} \to (V/U)^{**}, \quad [x]_{U^{**}} \mapsto \pi^{**}(x),$$

where π is the natural quotient map $V \twoheadrightarrow V/U$.

Dual Spaces and Galois Correspondence

Definition. (Kernels and Annulateurs). For convenience, let U_i denote subspaces of V, and B_i subspaces of V^*.

- $\operatorname{ann} U := f \in V^* \mid f(U) = 0$, a subspace of V^*;
- $\ker B := u \in V \mid f(u) = 0\ (\forall f \in B) = \bigcap_{f \in B} \ker f$, a subspace of V.

These notions are defined for subsets in general. By the closure property:

- $\operatorname{ann}(X) = \operatorname{ann}(\operatorname{span}(X))$, for $X \subseteq V$;
- $\ker(S) = \ker(\operatorname{span}(S))$, for $S \subseteq V^*$.

Remark. $\ker f$ and $\ker f$ are identical.

Exercise. Show that for any finite-dimensional vector space V, the map $\operatorname{ann} : \text{Subspaces of } V \to \text{Subspaces of } V^*$ is a bijection with inverse map $\ker$.

Proposition. One has $\operatorname{ann}(\ker B) \supseteq B$. Equality does not generally hold.

Proof. The formula $\operatorname{ann}(\ker B) \supseteq B$ involves logic more than algebra. For a counterexample, take V as the space of continuous functions over $\mathbb{R}$, and $B = \operatorname{span}(\{\delta_r\}_{r \in \mathbb{Q}})$.

Exercise. (Galois Connection)

1. $\operatorname{ann}(\ker B) \supseteq B$,
2. $\ker(\operatorname{ann} U) \supseteq U$,
3. $\operatorname{ann}(\ker(\operatorname{ann} B)) = \operatorname{ann}(B)$,
4. $\ker(\operatorname{ann}(\ker B)) = \ker B$.

The pair $(\ker, \operatorname{ann})$ defines a Galois correspondence.

Example. For a set map $f : X \to Y$, the image and preimage maps (f_*, f^*) form a Galois connection between $\text{Subset}(X)$ and $\text{Subset}(Y)$.

Definition. Let $Op(\mathbb{R}^2)$ denote the set of open subsets of $\mathbb{R}^2$, and $Cl(\mathbb{R}^2)$ the set of closed subsets. Define:

1. $k : Op(\mathbb{R}^2) \to Cl(\mathbb{R}^2), \quad U \mapsto \overline{U}$ (closure);
2. $i : Cl(\mathbb{R}^2) \to Op(\mathbb{R}^2), \quad K \mapsto K^\circ$ (interior).

Exercise. Show that $ikik = ik$ and $kiki = ki$; hence, (k, i) form a Galois connection.

Exercise. Given $X \subseteq \mathbb{R}^2$, show that at most seven distinct sets (including X) can be obtained by applying k and i repeatedly.

Exercise. Show that one can obtain at most 14 distinct sets (including X itself) by applying i and $(-)^c$ repeatedly.

The Yoga of Dual Spaces

Theorem. In the following table, FD denotes finite-dimensional cases, G denotes general cases, and HB refers to cases where the Hahn-Banach Axiom is assumed.

For a linear map $\varphi : U \to V$, define the dual map as

$$\varphi^* : V^* \to U^*, \quad [V \to \mathbb{F},\ v \mapsto f(v)] \mapsto [U \to \mathbb{F},\ u \mapsto f(\varphi(u))].$$

No.	Formula	FD	G	HB
G1	$\operatorname{ann}(\ker B) \overset{?}{=} B$	$=$	$\supseteq$	$\supseteq$
G2	$\ker(\operatorname{ann} V) \overset{?}{=} V$	$=$	$\supset$	$=$
E1	$\operatorname{ann}(V_1 + V_2) \overset{?}{=} \operatorname{ann}(V_1) \cap \operatorname{ann}(V_2)$	$=$	$=$	$=$
E2	$\operatorname{ann}(V_1 \cap V_2) \overset{?}{=} \operatorname{ann}(V_1) + \operatorname{ann}(V_2)$	$=$	$\supseteq$	$=$
E3	$\ker(B_1 + B_2) \overset{?}{=} \ker B_1 \cap \ker B_2$	$=$	$=$	$=$
E4	$\ker(B_1 \cap B_2) = \ker B_1 + \ker B_2$	$=$	$\supseteq$	$\supseteq$
Z1	$\ker \varphi^* \overset{?}{=} \operatorname{ann}(\operatorname{im} \varphi)$	$=$	$=$	$=$
Z2	$\ker \varphi \overset{?}{=} \ker(\operatorname{im} \varphi^*)$	$=$	$\subseteq$	$=$
Z3	$\operatorname{im} \varphi^* \overset{?}{=} \operatorname{ann}(\ker \varphi)$	$=$	$\subseteq$	$=$
Z3'	$\operatorname{im} \pi^* \overset{?}{=} \operatorname{ann}(\ker \pi)$, with π surj.	$=$	$=$	$=$
Z4	$\operatorname{im} \varphi = \ker(\ker \varphi^*)$	$=$	$\subseteq$	$=$
Z5	If f is surj., then f^* is inj.	true	true	true
Z6	If f is inj., then f^* is surj.	true	NaN	true

No.	Formula	FD	G	HB
L1	$S^* \xrightarrow{?} V^* / \operatorname{ann}(S)$	$\simeq$	$\hookleftarrow$	$\simeq$
L2	$(V/S)^* \xrightarrow{?} \operatorname{ann}(S)$	$\simeq$	$\simeq$	$\simeq$

Remark. When the general case (G) coincides with the HB case, the proposition is provable without additional axioms. When the FD case differs from the G case, the FD part is proved using dimension arguments. Nonexamples disprove the HB part of statements G1 and E4.

- G1-G2 come from Galois connection,
- E1-E4 are elementary,
- Z1-Z6 are taken from the textbook of 张贤科,
- L1-L2 are taken from the textbook of 黎景辉等.